Forgetting a Simple Root of a Binary Cubic

A coordinate-free model for a noninjective Keller map in dimension three

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Abstract

We describe intrinsically a polynomial map $F\colon \mathbb{A}^3_{\mathbb{C}} \to \mathbb{A}^3_{\mathbb{C}}$ with constant Jacobian determinant -2 and geometric degree three. Let W be a two-dimensional complex vector space, let H be a tangential affine slice of the space of binary cubics, and mark one simple projective root of a cubic in H. The resulting simple-root incidence variety is, exceptionally, isomorphic to $\mathbb{A}^3$. Under this identification, F is the morphism which forgets the marked root. Resultant normalization gives a coordinate-free proof of etaleness; the three roots of a general cubic explain the generic degree. We prove the affine-space identification, recover the explicit polynomial formula, and interpret the discriminant as the precise locus of nonproperness.

1 The intrinsic construction

Throughout, varieties are over $\mathbb{C}$. Let W be a two-dimensional vector space and fix a nonzero alternating form $\omega \in \bigwedge^2 W^\vee$. Choose a tensor $\tau \in \operatorname{Sym}^3 W$ of multiplicity type $(2,1)$: after extension to a basis it has the form $u^2 v$ with u, v independent. Equivalently, $[\tau]$ lies on the tangent developable of the Veronese cubic

$$\nu_3\colon \mathbb{P}(W) \longrightarrow \mathbb{P}(\operatorname{Sym}^3 W)$$

but not on the Veronese curve itself. Rescale τ once and for all, and put

$$H_\tau = \left\{ C \in \operatorname{Sym}^3 W^\vee : \langle \tau, C \rangle = 1 \right\}.$$

This is an affine hyperplane in a four-dimensional vector space, hence $H_\tau \simeq \mathbb{A}^3$. Consider the open simple-root incidence

$$X_\tau = \{ (C, [r]) \in H_\tau \times \mathbb{P}(W) : C(r) = 0 \text{ and } \mathrm{d}C_r \neq 0 \}.$$

Our main statement is the following.

Theorem 1.1 (Marked-root model)**.** *The variety X_τ is isomorphic to $\mathbb{A}^3$. Under this identification, the forgetful morphism*

$$\pi\colon X_\tau \longrightarrow H_\tau, \qquad (C, [r]) \longmapsto C,$$

is an etale polynomial map of affine three-spaces, generically of degree three. Its fiber over C is naturally the reduced set of simple projective roots of C.

Remark 1.2. The construction is coordinate-free but not choice-free. Its intrinsic data are (W, ω, τ) together with the chosen nonzero affine levels. The identifications of X_τ and H_τ with $\mathbb{A}^3$ are not canonical. Accordingly, the statement that π is etale is intrinsic, whereas the numerical value -2 of its Jacobian depends on the chosen affine volume coordinates.

The word "exceptionally" belongs to the assertion $X_\tau \simeq \mathbb{A}^3$. The target is visibly affine three-space, but the same need not be expected of a simple-root incidence for an arbitrary slice. The type-$(2,1)$ tensor is the special geometric datum behind the construction.

2 Resultant gauge and marked roots

The volume form gives an intrinsic scalar normalization of the linear–quadratic resultant. For $\ell \in W^\vee$, let $v_\ell \in W$ be determined by

$$\iota_{v_\ell}\omega = \ell.$$

For $q \in \operatorname{Sym}^2 W^\vee$, define

$$\operatorname{Res}_\omega(\ell, q) = q(v_\ell).$$

In an ω-unimodular basis this is the classical binary resultant of a linear and a quadratic form. In particular, it is nonzero precisely when ℓ and q are coprime. Define the normalized factor space

$$Y_{\omega,\tau} = \left\{(\ell, q) \in W^\vee \times \operatorname{Sym}^2 W^\vee : \operatorname{Res}_\omega(\ell, q) = 1, \quad \langle \tau, \ell q\rangle = 1\right\}. \tag{1}$$

Lemma 2.1 (Resultant gauge)**.** *Multiplication and passage to the root line induce an isomorphism*

$$Y_{\omega,\tau} \xrightarrow{\sim} X_\tau, \qquad (\ell, q) \longmapsto (\ell q, \mathbb{P}(\ker \ell)).$$

Proof. A marked root of C is a factorization $C = \ell q$, unique up to

$$(\ell, q) \longmapsto (s\ell, s^{-1}q), \qquad s \in \mathbb{G}_{\mathrm{m}}.$$

The root is simple if and only if ℓ and q are coprime. Moreover,

$$\operatorname{Res}_\omega(s\ell, s^{-1}q) = s \operatorname{Res}_\omega(\ell, q).$$

Thus every coprime scaling orbit has a unique resultant-one representative. Because the normalizing scalar is the inverse of a regular nowhere-zero function, this construction is algebraic in both directions. □

The weight one in the last formula is decisive: multiplication has a $\mathbb{G}_{\mathrm{m}}$-gauge, and the resultant cuts each coprime orbit exactly once.

There is also a root-gradient formulation. If $C = \ell q$ and $r = v_\ell$, then

$$\mathrm{d}C_r = q(r)\ell = \operatorname{Res}_\omega(\ell, q)\, \iota_r\omega.$$

Consequently $Y_{\omega,\tau}$ is equivalently the space of pairs

$$(C, r) \in H_\tau \times (W \setminus \{0\}), \qquad \mathrm{d}C_r = \iota_r\omega. \tag{2}$$

Euler's identity gives $3C(r) = \mathrm{d}C_r(r) = 0$, while the right-hand side is a nonzero covector. Thus the equation simultaneously says that r is a root and that it is simple. Conversely, every simple projective root has a unique representative satisfying (2).

3 A coordinate-free proof of etaleness

Consider the five-dimensional polynomial map

$$\Psi\colon W^\vee \times \operatorname{Sym}^2 W^\vee \longrightarrow \operatorname{Sym}^3 W^\vee \times \mathbb{A}^1, \qquad (\ell, q) \longmapsto (\ell q, \operatorname{Res}_\omega(\ell, q)).$$

Proposition 3.1. *The morphism Ψ is etale wherever $\operatorname{Res}_\omega(\ell, q) \neq 0$. Consequently, $\pi\colon X_\tau \to H_\tau$ is etale.*

Proof. Suppose that ℓ and q are coprime. If

$$\dot\ell q + \ell \dot q = 0,$$

then unique factorization implies that ℓ divides $\dot\ell$. Degree considerations give $(\dot\ell, \dot q) = a(\ell, -q)$ for some $a \in \mathbb{C}$. Hence the kernel of the differential of multiplication is exactly the infinitesimal gauge direction.

The resultant has gauge weight one, so

$$\mathrm{d}\operatorname{Res}_\omega\big|_{(\ell,q)}(\ell, -q) = \operatorname{Res}_\omega(\ell, q) \neq 0.$$

Therefore $\mathrm{d}\Psi$ is injective, hence an isomorphism between tangent spaces of dimension five. Finally,

$$Y_{\omega,\tau} = \Psi^{-1}(H_\tau \times \{1\}),$$

and its map to H_τ is the base change of Ψ along $H_\tau \times \{1\}$. Etaleness is preserved by base change, and Lemma 2.1 identifies this map with π. □

In coordinates the same argument is the relative-invariant identity

$$\det \mathrm{d}\big(\operatorname{coeff}(\ell q), \operatorname{Res}_\omega(\ell, q)\big) = \pm \operatorname{Res}_\omega(\ell, q)^2.$$

The differential proof explains the cancellation without expanding a 5×5 determinant.

4 The affine-space miracle

It remains to prove the substantive assertion $X_\tau \simeq \mathbb{A}^3$. Choose an adapted basis T, S of $W^\vee$, normalized so that $\omega = T \wedge S$ and $\langle \tau, C\rangle$ extracts the coefficient of $T^2 S$. Write

$$\ell = xT + \beta S, \qquad q = \gamma T^2 + \delta TS + \epsilon S^2.$$

Then (1) becomes the two equations

$$x\delta + \beta\gamma = 1, \qquad x^2\epsilon - x\beta\delta + \beta^2\gamma = 1. \tag{3}$$

Proposition 4.1 (Global affine chart)**.** *The map $\chi\colon \mathbb{A}^3_{x,y,z} \to Y_{\omega,\tau}$ defined by*

$$\begin{aligned}
\beta &= 1 + xy,\\
\gamma &= \frac{2 - 3xy + x^2 z}{2},\\
\delta &= -\frac{xz - y + x^2 yz - 3xy^2}{2},\\
\epsilon &= -z(1 + xy)^2 + 4y^2 + 3xy^3
\end{aligned}$$

is an isomorphism. Its polynomial inverse is

$$x = x, \qquad y = 2\beta\delta - x\epsilon, \qquad z = 4y\delta + 2\gamma y^2 - \epsilon. \tag{4}$$

Proof. Substitution verifies both equations in (3). Conversely, the two equations of the slice imply

$$\beta - 1 = x(2\beta\delta - x\epsilon) = xy,$$

so $\beta = 1 + xy$. Substitution into the first slice equation, followed by the definition of z in (4), gives

$$2\gamma = 2 - 3xy + x^2 z$$

and then

$$2\delta = y(1 + 3xy) - xz(1 + xy).$$

Finally the formula defining z gives

$$\epsilon = -z(1 + xy)^2 + y^2(4 + 3xy).$$

These are precisely the displayed coordinate functions. Both compositions are therefore the identity, including on $x = 0$; no divisor has been removed. □

This completes the proof that both source and target of π are affine three-spaces. Together with Proposition 3.1, it proves the etaleness assertion of Theorem 1.1.

5 The explicit Keller map

Write a cubic in H_τ as

$$C = c_3 T^3 + T^2 S + c_1 T S^2 + c_0 S^3$$

and use affine target coordinates $(2c_3, 2c_1, c_0)$. Multiplication $C = \ell q$, followed by the chart of Proposition 4.1, yields $G = (G_1, G_2, G_3)$ with

$$\begin{aligned} G_1 &= x^3 z - 3x^2 y + 2x, \\ G_2 &= -3x^3 y^2 z + 9x^2 y^3 - 6x^2 yz + 12xy^2 - 3xz + y, \\ G_3 &= -x^3 y^3 z + 3x^2 y^4 - 3x^2 y^2 z + 7xy^3 - 3xyz + 4y^2 - z. \end{aligned}$$

Because G is etale, its Jacobian determinant is a unit of $\mathbb{C}[x, y, z]$, hence a nonzero constant. Its linear part at the origin is

$$(x, y, z) \longmapsto (2x, y, -z),$$

and therefore

$$\boxed{\det JG = -2.}$$

The map in the original coordinate order is

$$\begin{aligned} F_1 &= (1 + xy)^3 z + y^2(1 + xy)(4 + 3xy), \\ F_2 &= y + 3x(1 + xy)^2 z + 3xy^2(4 + 3xy), \\ F_3 &= 2x - 3x^2 y - x^3 z. \end{aligned}$$

Indeed, if $S_0(x, y, z) = (x, y, -z)$ and $R_0(a, b, c) = (c, b, a)$, then

$$G = R_0 \circ F \circ S_0.$$

Thus F and G are the same etale morphism in different affine trivializations.

There is an equally compact normalized-root certificate. For $F(x,y,z)=(A,B,C)$ define

$$Q_{A,B,C}(U,V)=CU^3-2U^2V+BUV^2-2AV^3$$

and set $r=(U,V)=(1+xy,x)$. Directly,

$$Q_{A,B,C}(r)=0,\qquad (Q_{A,B,C})_U(r)=2V,\qquad (Q_{A,B,C})_V(r)=-2U.$$

For $\omega_0=\mathrm{d}U\wedge\mathrm{d}V$, this says

$$\mathrm{d}Q_{A,B,C}\big|_r=-2\,\iota_r\omega_0.$$

It is the same marked-root normalization as (2), after the harmless source and target rescalings relating F and G.

6 Fibers, discriminant, and infinity

The fiber description is immediate from the moduli interpretation.

Proposition 6.1. *For $C\in H_\tau$, the geometric fiber $\pi^{-1}(C)$ is the reduced set of simple projective roots of C. Its cardinality is*

$$\#\pi^{-1}(C)=\begin{cases}3, & C \text{ has three simple roots},\\ 1, & C \text{ has a double root and a simple root},\\ 0, & C \text{ has a triple root.}\end{cases}$$

In particular, π has geometric degree three and is not injective.

For instance, $C_0=TS(T+S)$ lies in H_τ and has three possible marked linear factors. In the coordinates of G these give

$$G(-1,2,-8)=G(0,2,16)=G(1,-1,-5)=(0,2,0).$$

Equivalently, in the original coordinates,

$$\begin{aligned}F\left(0,0,-\frac14\right)&=F\left(1,-\frac32,\frac{13}{2}\right)\\&=F\left(-1,\frac32,\frac{13}{2}\right)=\left(-\frac14,0,0\right).\end{aligned}$$

To see the boundary mechanism, introduce the full incidence

$$\overline{X}_\tau=\{(C,[r])\in H_\tau\times\mathbb{P}(W):C(r)=0\}.$$

Its projection to H_τ is finite flat of degree three. Its ramification locus consists of repeated marked roots, and its branch locus is the binary cubic discriminant. The source X_τ is obtained by deleting precisely that ramification locus upstairs. Over the complement of the discriminant, no point is deleted and π is finite etale. Over a double-root cubic the length-two ramified point is removed while the simple-root point remains; at a triple-root cubic every point is removed.

In the affine chart $X_\tau\simeq\mathbb{A}^3$, the deleted points appear at infinity. Thus π is etale and quasi-finite but not finite or proper. It should not be called a topological three-sheeted covering. Its generic monodromy is the ordinary S_3-monodromy of the roots of a cubic.

For the original target coordinates (a, b, c), the discriminant is

$$\Delta(a, b, c) = b^2 - cb^3 - 16a + 18abc - 27a^2c^2.$$

The nonproperness set is exactly $V(\Delta)$. Triple-root cubics form the omitted curve

$$(a, b, c) = \left(\frac{\rho^2}{3}, 2\rho, \frac{2}{3\rho}\right), \qquad \rho \in \mathbb{C}^*.$$

This is the geometric reason the map can be everywhere locally invertible without being globally invertible: its missing inverse branches do not ramify at finite source points; they escape through the boundary of the affine chart.

7 Conclusion

The coordinate formula conceals a short moduli problem:

binary cubic in a tangential affine slice + one normalized simple root $\longmapsto$ the cubic.

The resultant-one condition removes the factor-rescaling gauge, and its weight one proves etaleness. A general cubic has three roots, which explains the degree. The exceptional global chart $X_\tau \simeq \mathbb{A}^3$ converts this forgetful morphism into a polynomial self-map of affine three-space. The discriminant records exactly where inverse branches leave that chart. In this form, constant Jacobian, noninjectivity, and nonproperness are three aspects of the same marked-root geometry.

References

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